

EXERCISE 01: SQL FOUNDATIONS

SQL FUNDAMENTALS: SELECT & FILTERING

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01 SELECTING & SORTING

1. Retrieve all columns from the employees table.

SELECT *

FROM employees;

Output:

ID	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	ID	70000	2020-01-10	JHB
2	Brian	Lee	HR	45000	2019-03-12	CPT
3	Cathy	Zulu	finance	65000	2018-07-18	DUR
4	David	McKoenig	Marketing	50000	2021-11-05	PTA
5	Eva	Naidoo	IT IT	72000	2022-09-30	JHB

2. Find all unique departments: Expected Outcome

SELECT DISTINCT department

FROM employees;

Output:

department
ID
HR
Finance
Marketing

3. Retrieve first and last names ordered by salary descending: Expected outcome: first_name, last_name

```
SELECT first_name,  
       last_name  
FROM   employees  
ORDER BY salary DESC;
```

Outcome:

First_name	Last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

04. Retrieve the top 3 highest-paid employees
expected outcome: id, first_name, last_name, salary.

```
SELECT id,  
       first_name,  
       last_name,  
       Salary  
FROM   employees  
ORDER BY salary DESC  
LIMIT 3;
```

Output:

id	first_name	Last_name	Salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

02 : Filtering with WHERE

5. find employees in the IT department.

Expected outcome : Id, first_name, last_name, department

```
SELECT id,  
       first_name,  
       last_name,  
       department
```

```
FROM employees
```

```
WHERE department = 'IT';
```

Output :

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6. Find employees in Finance ^{and} with Salary > 60000
~~Find employees in HR or Marketing.~~

Expected outcome : id, first_name, last_name, department, Salary

```
SELECT id,  
       first_name,  
       last_name,  
       department,  
       salary
```

```
FROM employees
```

```
WHERE department = 'Finance'
```

```
AND salary > 60000;
```

Output :

id	first_name	last_name	department	Salary
3	Cathy	Zulu	Finance	65000

7. Find employees in HR or Marketing
expected Outcome: id, first_name, last_name, department.

```
SELECT id,  
       first_name,  
       last_name,  
       department
```

```
FROM employees
```

```
WHERE department = 'HR'
```

```
OR department = 'Marketing'
```

Output:

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Moores	Marketing

8. Find employees not in IT.

expected Outcome: id, first_name, last_name, department

```
SELECT id,  
       first_name,  
       last_name,  
       department
```

```
FROM employees
```

```
WHERE NOT department = 'IT';
```

Output:

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Moores	Marketing

9. Find employees in IT, HR or Finance using **IN**

Expected outcome: id, first_name, last_name, department

```
SELECT id,  
       first_name,  
       last_name,  
       department  
FROM employees
```

```
WHERE department IN ('IT', 'HR', 'Finance');
```

Output:

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10 Find employees in IT with salary > 65000 and city Johannesburg

Expected column: id, first_name, last_name, department, salary, city.

```
SELECT id,  
       first_name,  
       last_name,  
       department,  
       salary,  
       city
```

```
FROM employees
```

```
WHERE department = 'IT'
```

```
AND salary > 65000
```

```
AND city = 'Johannesburg';
```

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg